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EXPERIMENT 04

Data modelling and form design

Aim

Design a normalized service-center data model and build forms with proper types, relationships and validation.

Procedure

1. Create forms: Service_Request, Service_Types, Technician_Profile, Spare_Part_Inventory, Service_Request_Technician_Assignment, Service_Request_Part_Usage. Use stable link names.

2. Use lookups: Assignment.Service Request -> Service_Request, Assignment.Technician Profile -> Technician_Profile, Part_Usage.Service Request -> Service_Request, Part_Usage.Spare Part Inventory -> Spare_Part_Inventory. One request can have many assignments and part usages (1:N); a technician or part is reused across requests.

3. Set mandatory fields, dropdown choices and duplicate rules (Technician ID, Part ID and Request ID unique). Keep Vehicle Registration as text.

4. Do not delete requests or technicians with history; clear the Active box instead (e.g. T004 David Kim is Active = false in your data).

5. On create, set Status to Requested and Request_Date to today on the server in a workflow, not only by disabling UI fields.

6. Load synthetic fixtures only (e.g. TN01AB1234 Test Owner A). Never import real owner data.

Implementation and verification

Form	Fields and types (from your application)
Service_Request	Request ID; Request Date (date-time); Owner Name; Phone Number (+91); Owner Email; Vehicle Registration; Vehicle Make; Vehicle Model; Vehicle Year (number); Service Type (lookup/dropdown from Service Types); Problem Description; Status and file fields as you configured them.
Service_Types	Service type names used by Service Request (e.g. General Service, Oil Change, Brake Service).
Technician_Profile	Technician ID; Technician Name; Skill Specialization; Availability Status (On Break, On Leave, Available, Unavailable); Active (decision box).
Spare_Part_Inventory	Part ID; Part Name; Quantity Available; Reorder Level; Last Updated (date-time).
Service_Request_Technician_Assignment	Service Request (lookup); Technician Profile (lookup); Assignment Date (date-time); Assignment Notes (multi-line).
Service_Request_Part_Usage	Service Request (lookup); Spare Part Inventory (lookup); Quantity Used (number); Usage Date (date-time); Usage Notes (multi-line).

Evidence and testing

E04: relationship diagram, field dictionary, form screenshots, test records. Try a duplicate Technician ID, missing Vehicle Registration and invalid email; expect rejection. Derived values (parts remaining, technicians busy) should be computed or updated by workflow, not left as drifting editable counts.

Application images and explanation

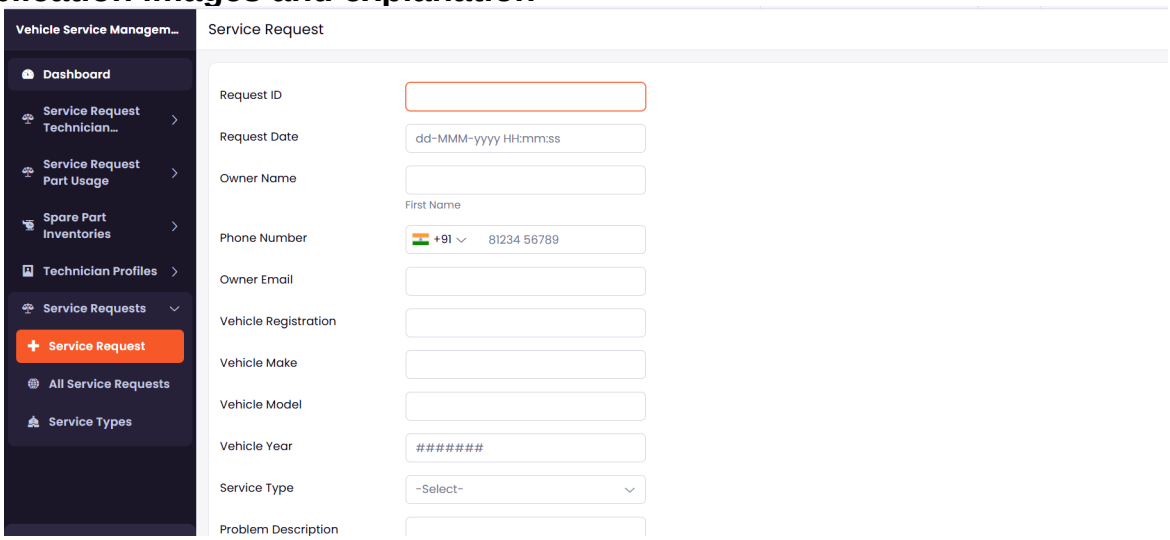


Figure 1: Service Request form

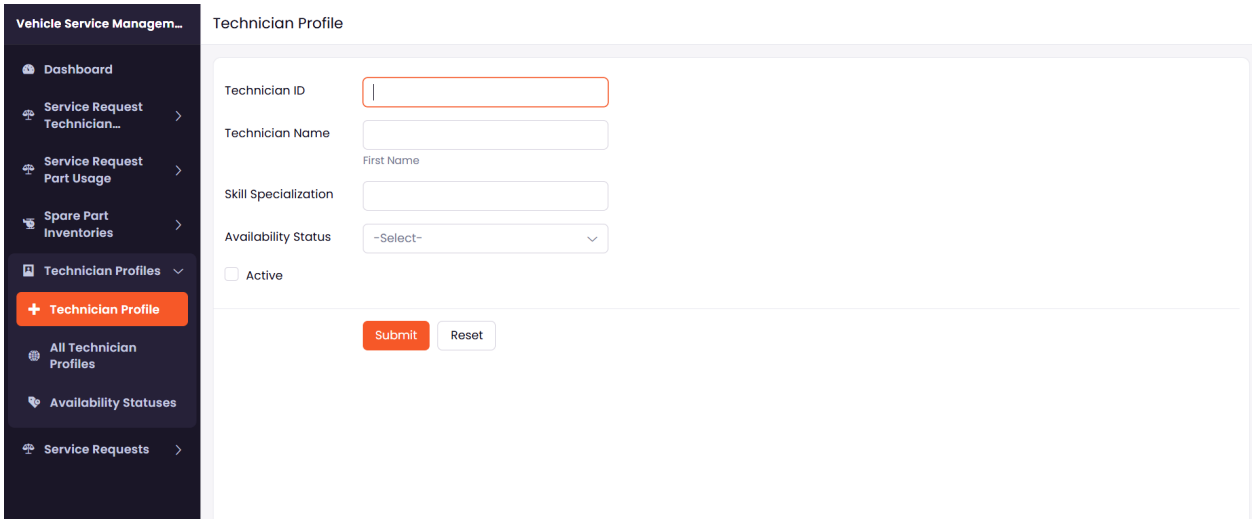


Figure 2: Technician Profile form

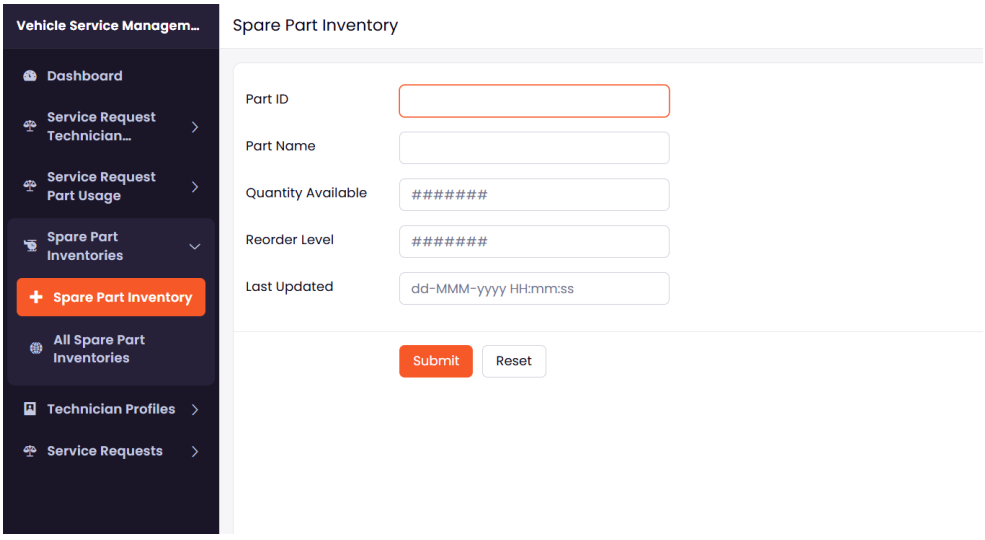


Figure 3: Spare Part Inventory form

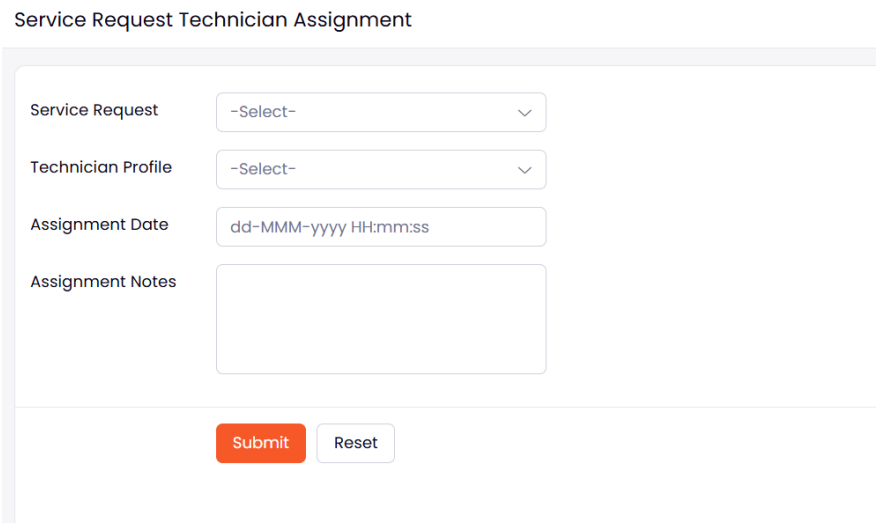


Figure 4: Service Request Technician Assignment form

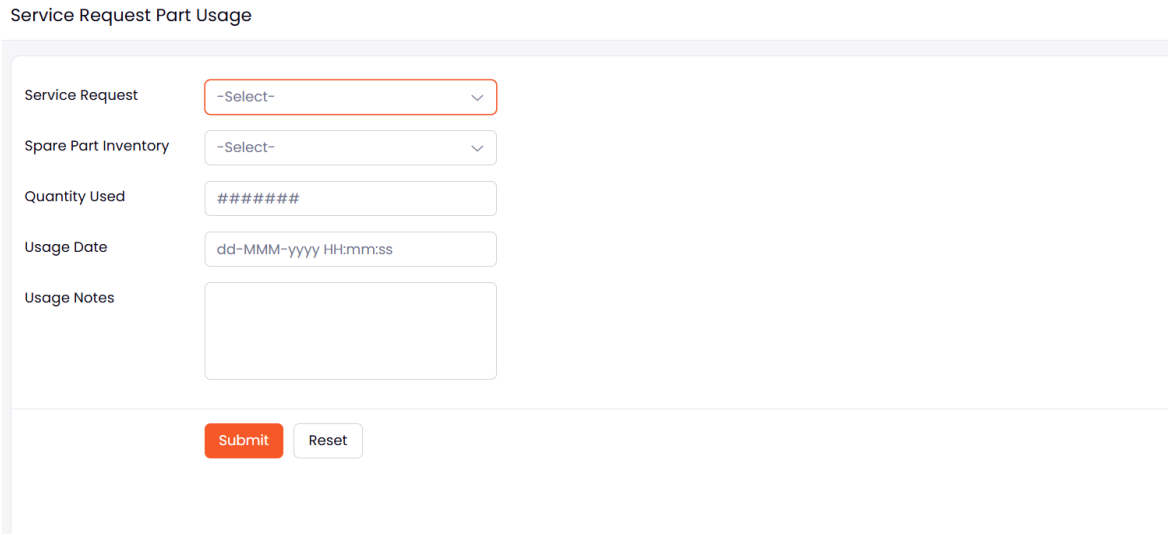
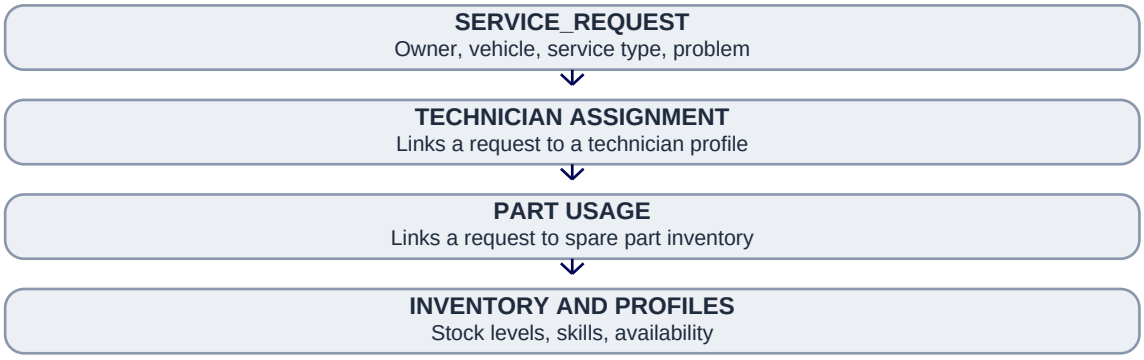


Figure 5: Service Request Part Usage form

Process diagram



Generated explanatory diagram. It is not a screenshot of an executed workflow.

Result

Expected result and verification criteria. All forms keep the intended relationships and constraints; invalid input is rejected; history is protected. No uniqueness or allocation guarantee is claimed from a lookup filter alone.